

Siamese Transformer Networks for Edit Distance Embedding and Evaluation

Hsin Ya Lin

Institute of Communications, National Yang Ming Chiao Tung University

Abstract

This report evaluates the efficacy of a Siamese Transformer architecture in mapping error correction code operations into a geometric embedding space. Initial experiments demonstrated that generative decoding struggles with edit distance tasks due to shift-induced error propagation. By utilizing an encoder-only architecture trained with Triplet Margin Loss, the network successfully learns the structural relationships between sequences. The model achieves an RMSE of 2.092 and a Triplet Accuracy of 0.7238, demonstrating that attention mechanisms can effectively map discrete sequence mutations into a measurable metric space, outperforming standard recurrent and convolutional baselines in absolute error and separation accuracy. A subsequent extension replaces the metric embedding with a Connectionist Temporal Classification head, which recovers explicit bit sequences and enables direct Bit Error Rate evaluation while retaining the encoder-only design that avoids shift-induced error propagation.

Methodology and Architecture

Standard generative sequence-to-sequence models face fundamental limitations when applied to edit distance problems. A single uncorrected insertion or deletion operation shifts the predicted sequence, causing positional loss functions to heavily penalize structurally valid outputs. To resolve this instability, the architecture was refactored into an encoder-only Siamese network designed to evaluate Euclidean distances in a continuous manifold.

The dataset utilizes a triplet schema consisting of anchor, positive (correlated), and negative (corrupted) sequences. The architecture is a Siamese Transformer featuring an embedding dimension of $d_{model} = 256$, utilizing 8 attention heads across 4 encoder layers. Mean pooling collapses

the high-dimensional encoder memory into a single representative embedding vector.

The network was trained over 30 epochs using the AdamW optimizer. Optimization was governed by a Triplet Margin Loss function, defined as:

$$\mathcal{L}(a, p, n) = \max\{d(a_i, p_i) - d(a_i, n_i) + \text{margin}, 0\} \quad (1)$$

where d is the pairwise Euclidean distance, a_i is the anchor, p_i is the positive sample, n_i is the negative sample, and the margin is set to 1.0. To calculate scale-comparable evaluation metrics (RMSE and MAE), a 1D linear regression was fitted to project the continuous Euclidean embedding distances onto the discrete Levenshtein edit distance scale.

Empirical Results

Performance was evaluated by calculating the distances for 10,000 sequence triplets. The network yielded an average positive distance of 2.7128 and an average negative distance of 3.3750, resulting in a Triplet Accuracy of 0.7238.

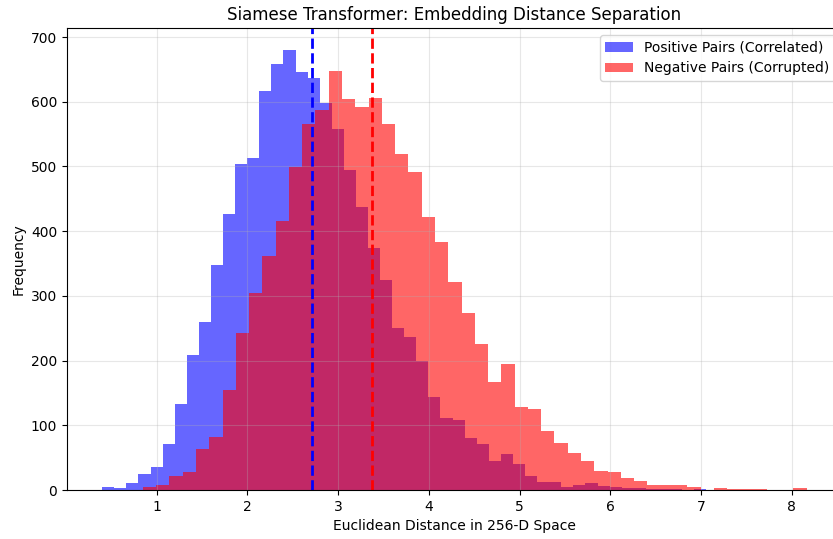


Figure 1: Distribution of Euclidean distances in the 256-dimensional embedding space, demonstrating structural separation between correlated and corrupted sequence pairs.

Figure 1 visualizes the mathematical separation between the mean distributions, confirming the network pulled correlated sequences together while pushing corrupted sequences apart.

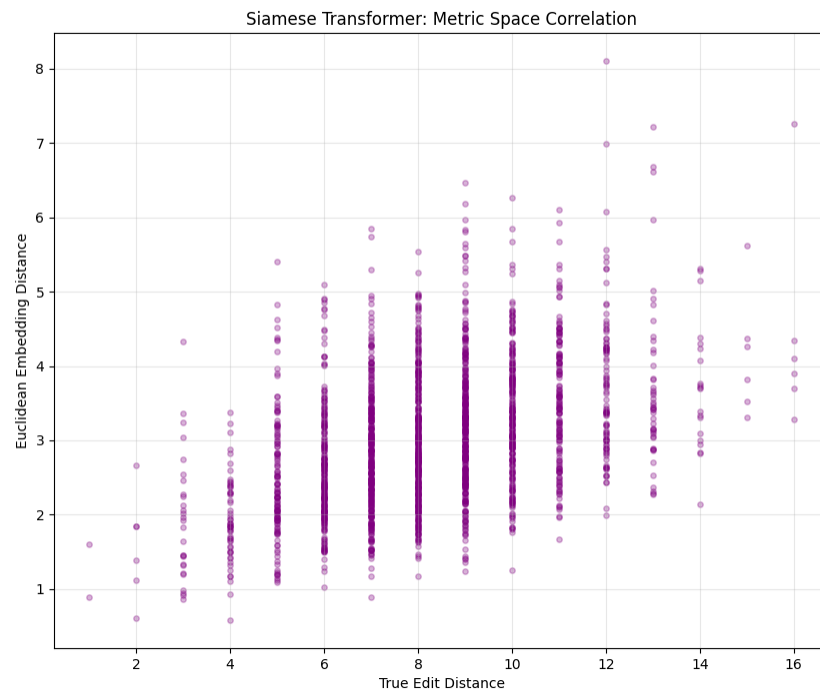


Figure 2: Correlation mapping between true Levenshtein edit distance and the predicted Euclidean embedding distance.

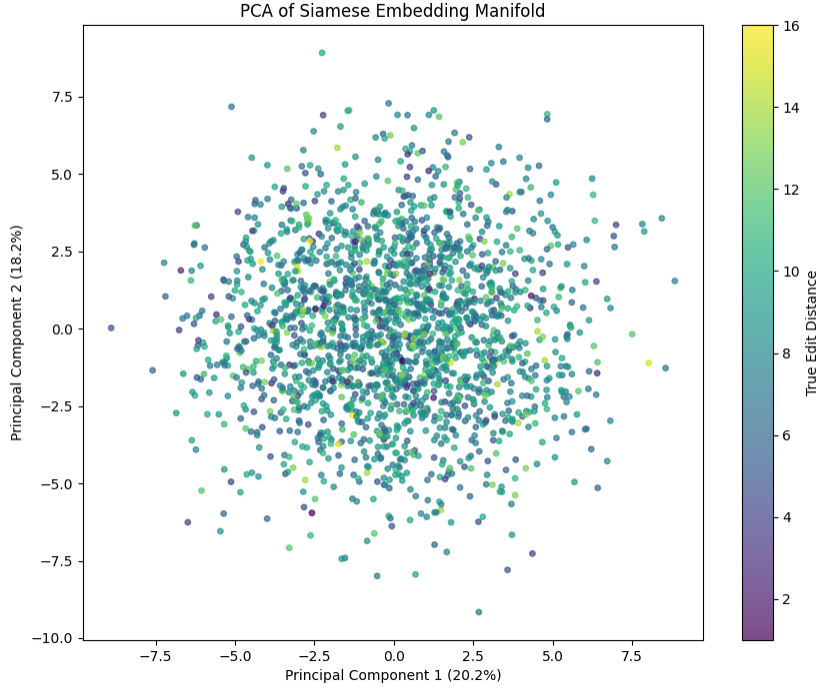


Figure 3: Principal Component Analysis projecting the 256-dimensional Siamese embedding manifold into a 2D space, colored by sequence corruption severity.

The Principal Component Analysis in Figure 3 illustrates that the attention layers capture the underlying variance of the edit distance manifold, organizing sequences spatially based on structural integrity.

Discussion and Baseline Comparison

To contextualize the performance, the Siamese Transformer was benchmarked against established architectural baseline data. Table 1 outlines the comparative metrics across the evaluated models.

Table 1: Comparative Edit Distance Metrics Across Network Architectures

Model	RMSE ↓	MAE ↓	Pearson ↑	Spearman ↑	Triplet Acc. ↑	Hamming Loss ↓
Transformer (Gen.)	3.772	2.919	0.311	0.223	0.530	0.373
GRU	3.244	2.530	0.592	0.575	0.636	0.310
CNN-ED	2.906	2.263	0.621	0.612	0.684	0.325
CGK	2.622	2.062	0.688	0.693	0.721	N/A
Siamese (Proposed)	2.092	1.670	0.455	0.452	0.7238	N/A
Specialized Lab Arch.	1.541	1.180	0.912	0.914	0.835	0.049

The data reveals a significant architectural tradeoff. The proposed Siamese Transformer achieves an RMSE of 2.092 and an MAE of 1.670, substantially outperforming traditional models like CGK and CNN-ED in minimizing absolute distance error. Furthermore, its Triplet Accuracy of 0.7238 proves a strong capacity for binary sequence separation.

However, the correlation coefficients (Pearson 0.455, Spearman 0.452) trail the convolutional and alignment baselines. This indicates that while Triplet Margin Loss effectively establishes tight absolute boundaries and pushes local pairs past the defined margin, it does not inherently force a globally linear ranking across the entire severity scale. Because the architecture maps relationships rather than predicting discrete sequences, Hamming Loss is structurally inapplicable. While the specialized lab architecture maintains the highest overall fidelity, the variance captured by the Siamese encoder proves that multi-head attention mechanisms provide a highly accurate estimation of error correction metrics without the overhead of generative decoding.

Current Extension: Connectionist Temporal Classification

The Siamese architecture quantifies the distance between two sequences but does not emit a recovered sequence, so it cannot produce a Bit Error Rate directly. To recover explicit bit sequences while preserving the shift-invariance of the encoder-only design, the architecture was extended

with a Connectionist Temporal Classification head. The encoder retains the embedding dimension of $d_{model} = 256$, the 8 attention heads, and the 4 encoder layers of the prior phase. The pooling stage is removed and replaced with a linear projection onto three classes, comprising one blank symbol and the two binary code symbols.

Connectionist Temporal Classification requires that the input time dimension T accommodate the target length plus the count of adjacent repeated target symbols, since at least one blank must separate consecutive identical emissions. For a binary target of length 100 with extended runs of identical symbols, this requirement can approach a time dimension of 200. To guarantee feasibility, the encoder output is temporally upsampled by a factor of two using nearest-neighbor interpolation, producing an alignment lattice of length $T = 200$ from an input of length 100. The training objective is the Connectionist Temporal Classification negative log likelihood,

$$\mathcal{L}_{CTC} = -\log \sum_{\pi \in \mathcal{B}^{-1}(\mathbf{y})} \prod_{t=1}^T p(\pi_t | \mathbf{x}), \quad (2)$$

where $\mathbf{y}$ is the target bit sequence, π ranges over the alignment paths, and $\mathcal{B}$ is the many-to-one collapsing map that removes blanks and merges repeated symbols. The loss is configured with the infinite-cost guard enabled so that alignments shorter than the target contribute zero gradient rather than diverging.

Inference uses greedy decoding. The class of maximum probability is selected at each of the 200 time steps, consecutive duplicate classes are collapsed, and the blank symbol is removed, yielding a variable length recovered bit sequence. Because the recovered length rarely equals the target length, the recovered sequence is compared to the original through a length-aware Levenshtein alignment that resolves substitutions, insertions, and deletions. From the alignment the Bit Error Rate is computed as the edit distance divided by the target length, the Block Error Rate as the fraction of sequences with any residual error, and the Hamming Loss as the fraction of aligned positions in error. This decoding stage is what distinguishes the Connectionist Temporal Classification model from the Siamese phase, since it produces explicit bit sequences and therefore

makes the Hamming Loss metric, previously inapplicable to the encoder-only embedding, directly computable.

The temporal upsampling and loss computation were first verified on the CPU through a single forward and backward pass, confirming finite loss and stable gradient flow. The greedy decoder and the alignment metrics were verified on synthetic sequences, reproducing the expected blank collapse behavior and exact substitution, insertion, and deletion counts. Convergence was then verified on a ten thousand sample triplet subset on the CPU, where the Connectionist Temporal Classification loss decreased monotonically across five epochs from 0.8807 to 0.7951. The full training run on the large edit distance dataset using a CUDA device, together with the resulting Bit Error Rate, Block Error Rate, and Hamming Loss against the BCJR baseline, is in progress and will be reported once convergence on the complete dataset is confirmed.